

Supplementary: Notation Reference

Supplementary: Notation Reference I

This supplement provides a comprehensive reference for the mathematical notation used throughout the Cognitive Integrity Framework (CIF) manuscript, including the eusocial and colony cognitive security extensions. Symbols are organized by domain, with cross-references to their formal definitions in the main text and supplements.

Adversary Model Notation I

Adversary Model Notation I

Symbol	Meaning	Defined In
Ω_k	Adversary class k (e.g., Ω_1 = External, Ω_5 = Systemic)	def:adversary-class
$\mathcal{R}$	Attack resource tuple $\langle R_C, R_K, R_A, R_P, R_{Co} \rangle$	def:resources
R_C	Computational resources (FLOPS-hours)	tab:resource-types
R_K	Knowledge resources (system understanding)	tab:resource-types
R_A	Access resources (available channels)	tab:resource-types
R_P	Persistence resources (temporal presence)	tab:resource-types
R_{Co}	Coordination resources (multi-party synchronization)	tab:resource-types
D_{score}	Detectability score of an attack	def:detectability
$\mathcal{C}_{\text{adv}}$	Adversarial capability set	def:capability-set
$\mathcal{K}_{\bullet}$	Adversary knowledge component of the class tuple; the subscript names its scope $(\mathcal{K}_{\text{public}}, \mathcal{K}_{\text{domain}}, \mathcal{K}_{\text{protocol}}, \mathcal{K}_{\text{internal}})$	sec:threat-model
$\mathcal{E}$	Communication edge set $\{(a_i, a_j) : \mathcal{C}(a_i, a_j) = 1\}$; the adversary-controlled subset is $\mathcal{E}_{\text{ctrl}} \subseteq \mathcal{E}$ (distinct from the stigmergic $\mathcal{E}$ below)	def:omega4
$\mathcal{A}_{\text{BIM}}$	Belief injection/manipulation attack class	sec:attack-taxonomy
$\mathcal{A}_{\text{BI}}$	Belief injection attack	thm:belief-injection

System Model Notation I

System Model Notation I

Symbol	Meaning	Defined In
$\mathcal{O}$	Multiagent operator tuple $\langle \mathcal{A}, \mathcal{C}, \mathcal{S}, \mathcal{P}, \Gamma \rangle$	def:multiagent-operator
$\mathcal{A}$	Set of agents $\{a_1, \dots, a_n\}$	tab:operator-components
a_i	Individual agent i	def:multiagent-operator
n	Number of agents	Throughout
$\mathcal{C}$	Communication adjacency matrix	tab:operator-components
$\mathcal{S}$	Shared global state	def:multiagent-operator
$\mathcal{P}$	Permission mapping	def:permission-layer
Γ	Coordination protocol	def:multiagent-operator
σ_i	Cognitive state of agent a_i : $\langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle$	def:cognitive-state
$\mathcal{B}_i$	Belief distribution of agent a_i : $\Phi \rightarrow [0, 1]$	tab:cognitive-components
$\mathcal{G}_i$	Goal set of agent a_i	tab:cognitive-components
$\mathcal{I}_i$	Intention set (committed actions)	tab:cognitive-components
$\mathcal{H}_i$	Interaction history	tab:cognitive-components
S^t	Global system state at time t	def:system-state
σ_i^t	Cognitive state of agent i at time t	def:cognitive-state
Φ	Set of propositions	sec:notation
ϕ, ψ	Individual propositions	sec:notation
$\mathcal{M}$	Message space (distinct from the set of marker types $\mathcal{M}$ of def:stigmergic-operator)	def:firewall
m	Individual message	def:firewall
Σ	OODA input alphabet $\mathcal{M} \cup \text{Events}$ (distinct from the stigmergic update function Σ of def:stigmergic-operator)	def:ooda-state

Trust Calculus Notation I

Trust Calculus Notation I

Symbol	Meaning	Defined In
$\mathcal{T}_{i \rightarrow j}$	Trust score from agent i to agent j	def:trust-function
$\mathcal{T}_{\text{base}} / T_{\text{base}}$	Base architectural trust (role-based)	tab:trust-components
$\mathcal{T}_{\text{rep}} / T_{\text{rep}}$	Reputation trust (historical accuracy)	tab:trust-components
$\mathcal{T}_{\text{ctx}} / T_{\text{ctx}}$	Contextual trust (task-specific)	tab:trust-components
α, β, γ	Trust component weights ($\alpha + \beta + \gamma = 1$)	eq:trust-function
δ	Trust decay factor ($\delta \in (0, 1)$)	def:trust-delegation
d	Delegation depth	def:trust-delegation
$\mathcal{T}^{\text{del}}$	Delegated trust value	def:trust-delegation
$\mathcal{T}^{\text{path}}$	Path trust value	def:path-trust
η_m	Modality reliability factor	def:modality-trust
η	Learning rate (reputation update)	Trust configuration
ρ	Penalty factor (failure penalty multiplier)	Trust configuration
$\otimes$	Trust delegation operator	def:trust-algebra
$\oplus$	Trust aggregation operator	def:trust-algebra

Defense Mechanism Notation I

Defense Mechanism Notation I

Symbol	Meaning	Defined In
$\mathcal{F}(m)$	Cognitive firewall classification function	def:firewall
D_{inj}	Injection detection score	def:firewall-rules
D_{sus}	Suspicious content score	def:firewall-rules
τ_1	Firewall reject threshold	eq:firewall-rules
τ_2	Firewall quarantine threshold	eq:firewall-rules
$\mathcal{B}_{\text{verified}}$	Set of verified beliefs	def:sandbox
$\mathcal{B}_{\text{provisional}}$	Set of provisional (sandboxed) beliefs	def:sandbox
$\pi(\phi)$	Provenance chain for belief ϕ	def:evidence
$V(\pi)$	Provenance verification function	sec:integrity-properties
$\mathcal{W}$	Set of canary beliefs (tripwires)	def:canary
ω_j	Individual canary belief	eq:canary-set
p_j^{exp}	Expected probability for canary j	eq:canary-set
ϵ_{drift}	Drift detection threshold	eq:tripwire-alert
$\mathcal{I}_{\text{inv}}$	Set of behavioral invariants	def:invariant-set
I_k	Individual invariant predicate	eq:invariant-set
κ	Corroboration threshold	sec:belief-update-rules
TTL	Time-to-live for provisional beliefs	Sandbox configuration

Detection & Analysis Notation I

Detection & Analysis Notation I

Symbol	Meaning	Defined In
S_{drift}	Drift score (belief change magnitude)	def:drift-score
D_{KL}	Kullback-Leibler divergence (drift detection)	def:drift-detection
w	Sliding window size	def:drift-detection
g_t	CUSUM cumulative statistic at time t	def:cusum-detector
ν	CUSUM allowance (reference value)	def:cusum-detector
h	CUSUM decision interval (alert threshold on g_t)	def:cusum-detector
λ	Max delta weight in drift scoring	eq:drift-score
λ	Chernoff tilting parameter, $\lambda \in [0, 1]$ (unrelated to the drift weight above and to the colonial-trust decay constant)	eq:chernoff
S_{dev}	Behavioral deviation score	def:deviation-score
f_k	Feature extractor function	eq:deviation-score
μ_k, σ_k	Feature mean and standard deviation	eq:deviation-score
AUC	Area Under the ROC Curve	def:auc
$\text{TPR}(\tau)$	True Positive Rate at threshold τ	eq:tpr
$\text{FPR}(\tau)$	False Positive Rate at threshold τ	eq:fpr
$\text{FNR}(\tau)$	False Negative Rate at threshold τ	eq:threshold-opt
S_{fused}	Fused detector score	def:score-fusion
D_{fused}	Fused detector decision	def:decision-fusion
$\text{taint}(\phi)$	Provenance tags for belief ϕ	def:taint-label
$\Lambda(m)$	Likelihood ratio $P_{\text{attack}}(m)/P_{\text{benign}}(m)$ (capital Λ ; unrelated to λ)	def:hyp-test

Consensus & Coordination Notation I

Consensus & Coordination Notation I

Symbol	Meaning	Defined In
q	Quorum threshold for consensus	def:quorum
f	Maximum number of Byzantine/compromised agents	thm:byzantine-req
$\mathcal{B}_{\text{consensus}}$	Consensus belief function	def:cog-byzantine
$\mathcal{D}$	Set of defense mechanisms	def:defense-composition
$\mathcal{D}_1 \circ \mathcal{D}_2$	Series defense composition	eq:series-comp
$\mathcal{D}_1 \parallel \mathcal{D}_2$	Parallel defense composition	eq:parallel-comp
P_{detect}	Detection probability	eq:series-detection
r_f	Firewall detection rate	thm:belief-injection
r_s	Sandbox verification rate	thm:belief-injection

Cost & Performance Notation I

Cost & Performance Notation I

Symbol	Meaning	Defined In
C_{total}	Total defense cost	def:defense-cost
C_{compute}	Computational cost	tab:cost-components
C_{latency}	Latency cost	tab:cost-components
C_{fp}	False positive cost	tab:cost-components
$C_{\text{FP}}, C_{\text{FN}}$	Cost of false positive / false negative	def:cost-threshold
B_{total}	Total defense benefit	def:defense-benefit
L_{CIF}	CIF latency overhead	thm:latency-overhead
L_d	Latency of defense d	eq:latency-budget
L_{max}	Maximum allowed latency	eq:latency-budget

Information & Complexity Notation I

Information & Complexity Notation I

Symbol	Meaning	Defined In
$H(\mathcal{A})$	Entropy of attack $\mathcal{A}$	thm:min-entropy
$I(D; \mathcal{A})$	Mutual information between detector and attack	def:detector-gain
C_{channel}	Channel capacity	thm:stealth-impact
$O(\cdot)$	Big-O complexity bound	sec:complexity-bounds
S_{total}	Total space complexity	eq:total-space
T_{msg}	Per-message processing time	eq:message-processing

Stigmergic & Colony Notation (Supplementary) I

Stigmergic & Colony Notation (Supplementary) I

Symbol	Meaning	Defined In
$\mathcal{O}_\Sigma$	Stigmergic operator tuple	def:stigmergic-operator
$\mathcal{E}$	Environmental state (markers/signals)	def:stigmergic-operator
Σ	Stigmergic update function (distinct from the OODA input alphabet Σ of def:ooda-state)	def:stigmergic-operator
$\mathcal{L}$	Set of locations	def:stigmergic-operator
$\mathcal{M}$	Set of marker types (distinct from the message space $\mathcal{M}$ of def:firewall)	def:stigmergic-operator
$\mathcal{N}$	Cyberphysical niche	def:cyberphysical-niche
$\mathcal{F}_c$	Emergent collective function	def:emergent-function
$\mathcal{T}_c$	Colonial trust function (environment-mediated)	def:colonial-trust
$\rho(m, l, t)$	Signal reliability at location l for marker m at time t	eq:colonial-trust
λ	Temporal decay constant (colonial trust)	eq:colonial-trust
Q_α	Cognitive quorum function for action α	def:cognitive-quorum
$\mathcal{A}_e$	Emergent attack	def:emergent-attack
CCS	Colony CogSec Score	def:cogsec-score
DR _c	Colony-level detection rate	eq:ccs
FPR _c	Colony-level false positive rate	eq:ccs

General Mathematical Notation I

General Mathematical Notation I

Symbol	Meaning	Usage
$P(\cdot)$	Probability measure	Throughout
$\mathbb{1}[\cdot]$	Indicator function	eq:decision-fusion
τ	Generic threshold parameter	Throughout
θ_{drift}	KL drift-detection threshold; distinct from the per-canary deviation tolerance ϵ_{drift}	eq:drift-detection
$\theta_{\text{total}}, \theta_{\text{step}}$	Cumulative and per-step drift thresholds in a progressive-drift attack	def:progressive-drift
θ (alarm)	Baseline drift alarm threshold	prop:drift-threshold
θ_i	$\mu_{\text{baseline}} + k \sigma_{\text{baseline}}$ Escalation threshold at stage i of the multi-stage detection pipeline	def:multi-stage
θ (fusion)	Learned parameters of the fusion network — not a threshold	eq:learned-fusion
r_θ	Fisher-Rao displacement corresponding to θ_{drift}	rem:geometric-bound
ϵ	Small constant (error rate, deviation)	Throughout
t	Time index	Throughout
$\models$	Satisfaction relation (state satisfies predicate)	eq:invariant-check
$\vdash$	Logical entailment	eq:consistency-def
$\perp$	Logical contradiction	eq:consistency-def
$\perp\!\!\!\perp$	Undecided / undefined	eq:cog-byzantine
$\checkmark$	Verification passed	tab:mc-results

CTL Temporal Logic Notation (Formal Verification) I

CTL Temporal Logic Notation (Formal Verification) I

Symbol Meaning		Defined In
AG	"Always globally" (CTL operator)	eq:ctl-safety
AF	"Always eventually" (CTL operator)	eq:ctl-liveness
EX	"Exists next" (CTL operator)	sec:temporal-properties
$\Rightarrow$	Logical implication	Throughout
$\leftrightarrow$	Logical biconditional	Throughout